

Immunogenicity and safety of reduced-dose versus full-dose mRNA COVID-19 booster (REDU-VAC): a single blind, randomized, noninferiority trial

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CONCLUSION: The full-dose induced higher initial anti-RBD IgG and neutralizing antibody titers at day 28 post-vaccination (non-inferiority not demonstrated). However, this difference waned over time, and the incidence of breakthrough infections was comparable between groups. Notably, the reduced-dose exhibited a more favorable reactogenicity profile. These findings suggest that reduced-dose boosting may offer a viable strategy to sustain immune protection while optimizing vaccine safety and distribution.

Regular booster doses for COVID-19 are necessary to strengthen and prolong the immune response following vaccination, ensuring continued protection against infection and severe disease. The potential benefits of reduced-dose booster vaccinations are multiple: viable solution to face limited vaccine supplies, less reactogenicity, and better acceptability. Here, we report the safety and immunogenicity of a fractional dose as a booster (third dose) of the BNT162b2 vaccine.

Methods

- Participant-blinded, randomized, phase 4, non-inferiority study investigating safety, reactogenicity and immunogenicity
- Follow-up of REDU-VAC study cohort where a reduced or full-dose of BNT162b2 was administered as prime doses
- Adults 18–55 years old, either COVID-19 previously infected (*P-Inf*) or infection naïve (*Naïve*), randomly assigned to receive a booster of 10µg (reduced-dose) or 30µg (full-dose) of BNT162b2
- Blood samplings before booster (D0), at day 28 (D28), and month 6 (M6)
- SARS-CoV-2 anti-receptor binding domain (RBD) specific IgG concentrations, neutralizing antibody titers against SARS-CoV-2 Wild type (WT) and omicron BA.1 variant assessed at the 3 time points. Avidity of WT RBD-specific IgG assessed at Days 0 & 28 in the naïve cohort
- Analyses using mixed-effect linear models
- Reduced-dose considered non-inferior to the full-dose if the lower limit of the two-sided 95% CI of the GMR was >0.67 (WHO recommended non-inferiority margin)
- COVID-19 status of participants was determined using three criteria: i) a self-reported, laboratory-confirmed SARS-CoV-2 infection, ii) a positive anti-N IgG test, and iii) an anti-RBD IgG ratio between 2 times ≥ 1

Results

- Titers at baseline were not significantly influenced by the prime dose concentration. On the contrary, previously infected participants had higher titers than naïve participants.
- Safety analysis**
 - No serious adverse event (AE). No difference in severity between 10µg and 30µg
 - Significantly lower number of participants reporting AE in the reduced dose than in the full-dose (41% vs. 78%, $P < 0.0001$) for both local and systemic AE

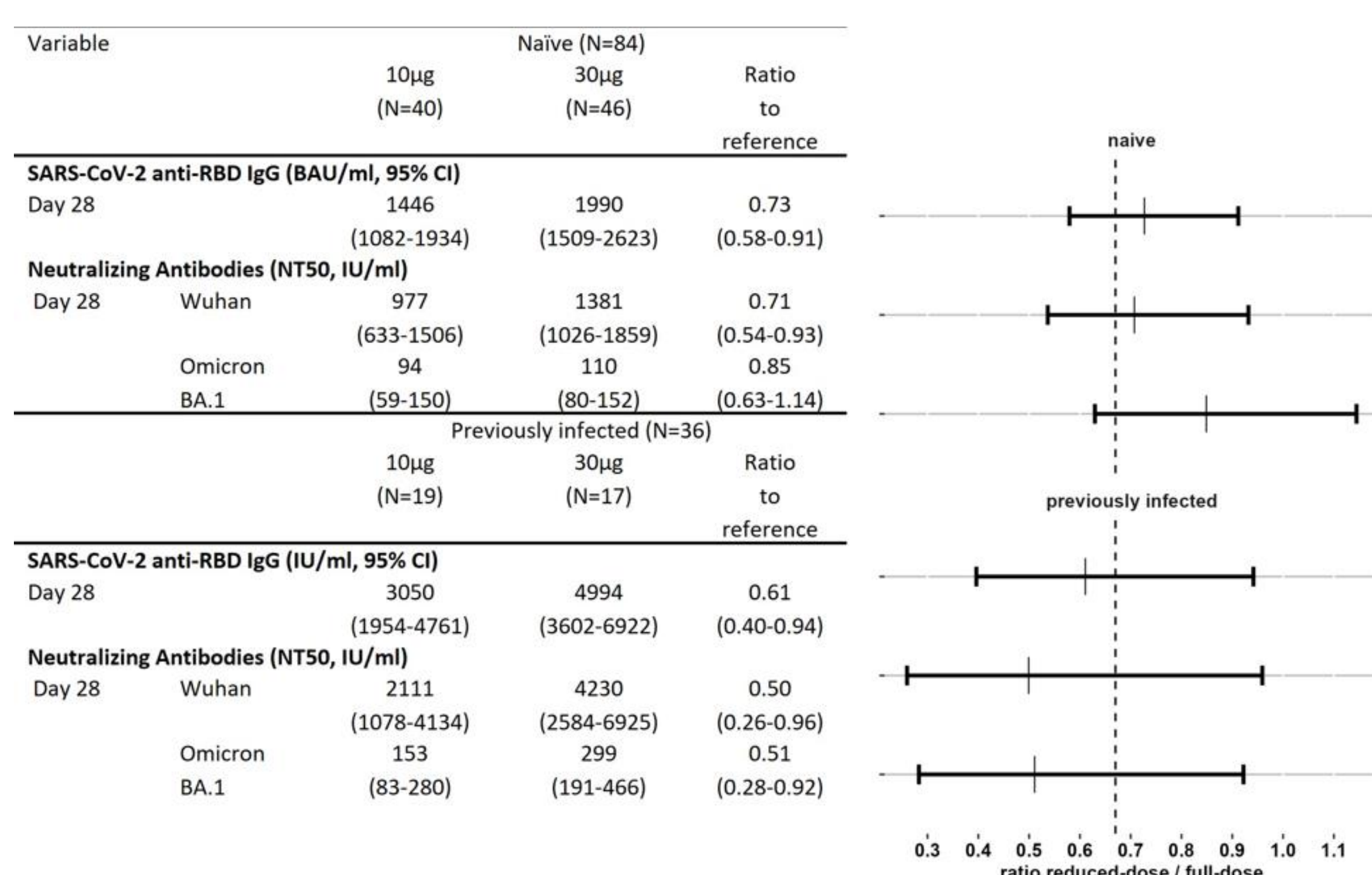


Fig 1. Immune responses by study arm at day 28 and non-inferiority analysis in the per-protocol cohorts. GMTs (95% CI) at day 28. GMRs (95% CI) adjusted with a linear mixed-effect model including sex, time between prime doses, time between dose 2 and dose 3, and SARS-CoV-2 anti-RBD IgG titer at baseline (Day 0) as fixed variables and location as random variable. Dashed line: WHO recommended non-inferiority margin of 0.67.

Table 1. Demographics, characteristics by study group

Characteristic		Naïve (N=84)		Previously infected (N=36)	
		10 µg (N=40)	30 µg (N=46)	10 µg (N=19)	30 µg (N=17)
Age (years, mean ± SD)		40.0 ± 7.6	38.2 ± 9.1	40.5 ± 10.5	38.4 ± 10.4
Sex (n, %)	Female	21 (53%)	25 (54%)	10 (53%)	11 (65%)
Ethnicity (n, %)	European	40 (100%)	44 (96%)	17 (90%)	17 (100%)
Time (days) between prime doses (mean ± SD)		27.0 ± 11.4	28.6 ± 9.4	27.8 ± 7.4	24.6 ± 6.2
Time (days) between dose 2 – dose 3 (mean ± SD)		186.3 ± 24.3	177.5 ± 26.5	184.3 ± 41.4	189.1 ± 25.9
Comorbidities (under control) (n, %)	Cardiovascular	1 (2%)	2 (4%)	0	1 (6%)
	Neurological	0	1 (2%)	0	0
	Oncological	0	0	0	1 (6%)
	HIV (PrRP)	0	0	0	1 (6%)

Immunogenicity (Fig. 2)

- Non-inferiority not demonstrated at D28 (Fig. 1) for both *P-inf* and *Naïve*
- Significant dose effect only demonstrated at D28 (30µg > 10µg) for both *P-inf* and *Naïve*, for all the immune variables
- Binding IgG waned after D28 if no BTI. *Naïve* had still higher titers at M6 than at D0, but it is not true for *P-inf*
- Neutralizing Ab against BA.1 were enhanced with the booster but still 30-40% had undetectable titers in the *Naïve* group, with no dose effect on this proportion
- The rate of breakthrough infections (BTI) after D28 was not significantly different between 10µg and 30µg (69% versus 62%, $p=0.56$).
- Immune titers were significantly boosted after BTI (at M6) compared to people without a BTI, in both *P-inf* and *Naïve* groups.

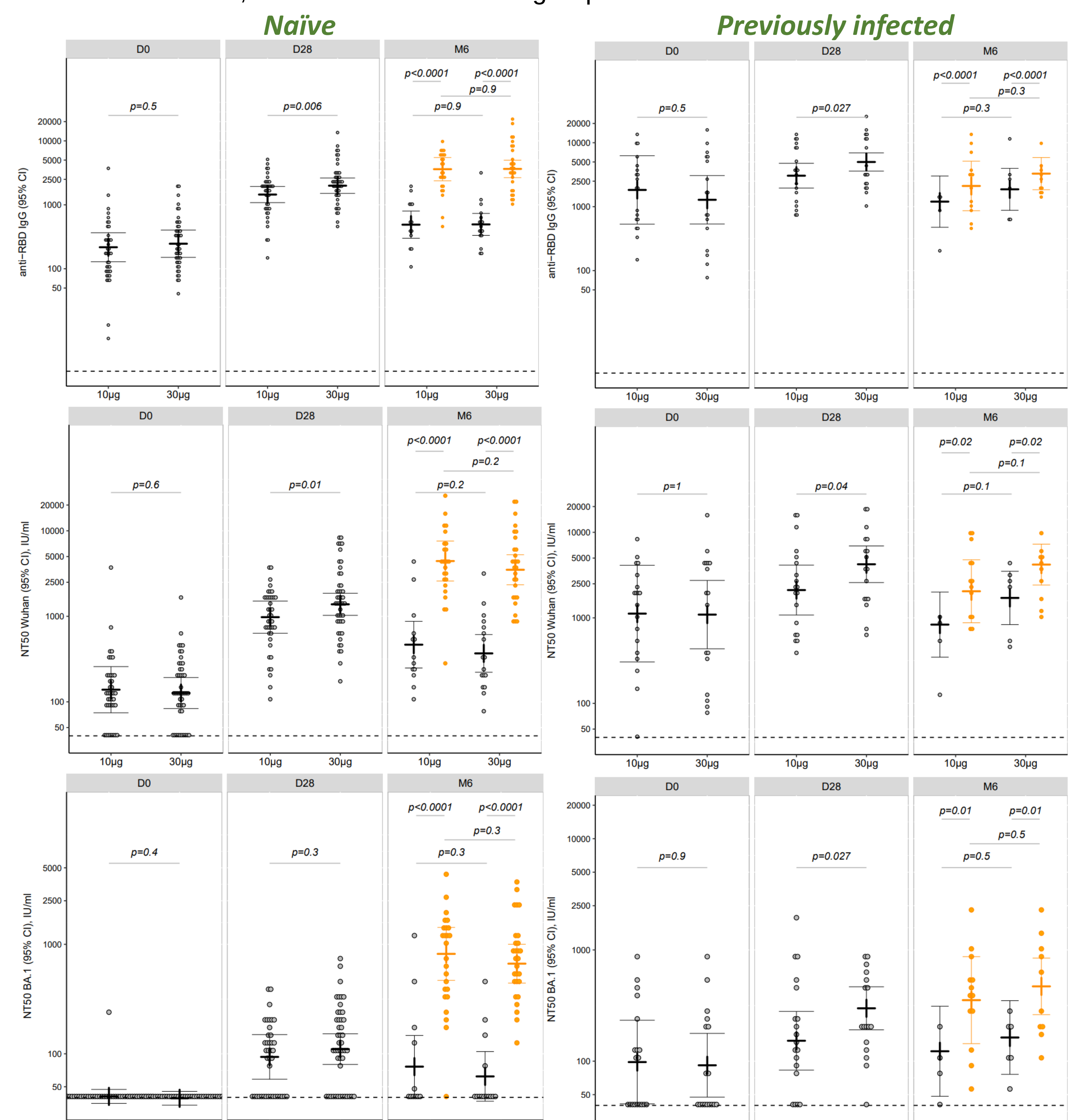


Fig 2. Kinetics of immune responses (anti-RBD IgG, NT50 Wuhan and NT50 BA.1 nAb) in naïve and previously infected groups, for both booster concentration: reduced-dose (10µg) and full-dose (30µg). Individual responses and geometric mean titers with 95% CI are shown. People having a BTI after D28 (orange points) are differentiated from those without a BTI (black points). Dashed line: Limit of Detection (LOD)

REFERENCES

- Pannus, P. et al., Safety and immunogenicity of a reduced dose of the BNT162b2 mRNA COVID-19 vaccine (REDUVAC): A single blind, randomized, non-inferiority trial, *PLOS Glob Public Health* 2022, 2(12): e0001308.

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